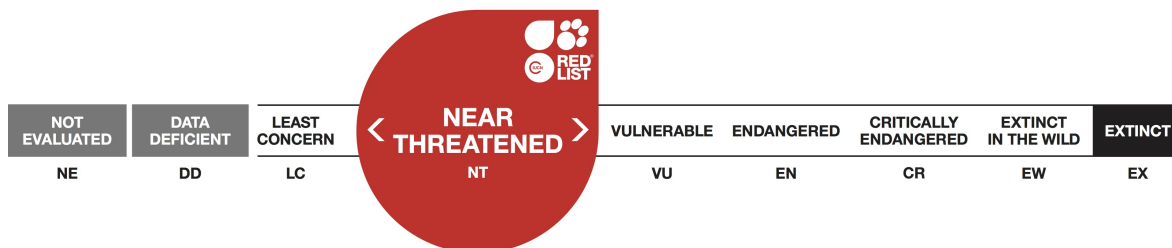


Phoebastria immutabilis, Laysan Albatross

Assessment by: BirdLife International



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Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Aves	Procellariiformes	Diomedidae

Taxon Name: *Phoebastria immutabilis* (Rothschild, 1893)

Synonym(s):

- *Diomedea immutabilis* ssp. *immutabilis* — Sibley and Monroe (1990, 1993)
- *Diomedea immutabilis* ssp. *immutabilis* — Stotz *et al.* (1996)

Common Name(s):

- English: Laysan Albatross
- Spanish: Albatros de Laysan

Taxonomic Source(s):

Robertson, C. J. R.; Nunn, G. B. 1998. Towards a new taxonomy for albatrosses. In: Robertson, G.; Gales, R. (ed.), *Albatross biology and conservation*, pp. 13-19. Surrey Beatty & Sons, Chipping Norton, Australia.

Identification Information:

81 cm. Small two-tone, gull-like albatross. Upperwings, mantle, back, upper rump and tail blackish-grey. Head, lower rump and underparts white. Blackish smudge around eye. Bill pinkish with darker tip. Black-and-white underwing pattern varies between individuals having narrow black margins and variable amounts of black in the underwing coverts. Juveniles very similar but for greyer bill and wholly dark upper rump. **Similar spp.** Easily separated from the wholly dark Black-footed Albatross *P. nigripes*. Adult Short-tailed Albatross *P. albatrus* has a white back.

Assessment Information

Red List Category & Criteria: Near Threatened [ver 3.1](#)

Year Published: 2018

Date Assessed: August 7, 2018

Justification:

This species has rebounded from declines in the late 1990s and early 2000s, perhaps because apparent changes in the breeding populations reflected large scale environmental conditions that affected the number of birds that returned to the colonies to nest rather than actual declines in the population. Given the difficulty of predicting long-term trends for such a long-lived species, and the number of documented threats and the uncertainty over their future effects, the species is precautionarily projected to undergo a moderately rapid population decline over three generations (84 years), and as such qualifies as Near Threatened (nearly qualifies for listing as threatened under criterion A4bd).

Previously Published Red List Assessments

2016 – Near Threatened (NT)

<http://dx.doi.org/10.2305/IUCN.UK.2016-3.RLTS.T22698365A93679937.en>

2016 – Near Threatened (NT)

2013 – Near Threatened (NT)

<http://dx.doi.org/10.2305/IUCN.UK.2013-2.RLTS.T22698365A49901920.en>

2012 – Near Threatened (NT)

2010 – Near Threatened (NT)

2008 – Vulnerable (VU)

2005 – Vulnerable (VU)

2004 – Vulnerable (VU)

2003 – Vulnerable (VU)

2000 – Lower Risk/least concern (LR/lc)

1994 – Lower Risk/least concern (LR/lc)

1988 – Lower Risk/least concern (LR/lc)

Geographic Range

Range Description:

Phoebastria immutabilis breeds at 16 sites (nine with populations of greater than 100 pairs), mostly in the Northwestern Hawaiian Islands (**U.S.A.**) and **U.S. Minor Outlying Islands**, with additional small colonies in **Japan** and **Mexico**. The population is estimated to be more than c. 800,000 breeding pairs, with the largest colony at Midway Atoll (666,044 pairs, [https://www.fws.gov/uploadedFiles/Region_1/NWRS/Zone_1/Midway_Atoll/Documents/Hatch%20Year%202015%20NR_FINAL\(4\).pdf](https://www.fws.gov/uploadedFiles/Region_1/NWRS/Zone_1/Midway_Atoll/Documents/Hatch%20Year%202015%20NR_FINAL(4).pdf)), followed by Laysan Island, both in the Northwestern Hawaiian Islands (Arata *et al.* 2009). Population sizes at monitored colonies increased between 1980 and 1995 but have never reached the densities observed prior to large-scale harvests for feathers in the early 1900s. Data indicated a 32% decline during 1992-2002 (3.2% per annum) of birds breeding on the Northwestern Hawaiian Islands, where 99% of the global population is found (Gilman and Freifeld 2003, USFWS data *per B. Flint 2003 pers comm.*), though data from 2004 and 2006 indicate that the breeding population then rebounded, and that the overall population trend for 1992-2005 is stable (Naughton *et al.* 2007). On Oahu, Hawaii the small population has increased 27% annually since 1991, and numbered 555 adults in 2015, due primarily to immigration with some local recruitment (Young *et al.* 2009a, VanderWerf and Young 2016). The island of Kauai contain ~1,000 birds scattered amongst more than a dozen sites with an increasing trend comparable to Oahu (Young *et al.* 2014). A population began nesting in Mexico in the 1980s and has been increasing since then. The current population is about 400 pairs at four sites (46 pairs on Clarion Island in 2002; Wanless *et al.* 2009), though this represents less than 0.1% of the global population (Arata *et al.* 2009). Breeding populations were extirpated from Wake and Johnston atolls (U.S.A.) and Minami Torishima (Japan) in the central Pacific. Ship-based observations, satellite tracking and fisheries bycatch reveal the wide distribution of *Phoebastria immutabilis* in the North Pacific, ranging from the northern Bering Sea to tropical waters in the South (10-15 degrees North) (Fisher and Fisher 1972, Fernandez *et al.* 2001, Hyrenbach *et al.* 2002, Shaffer *et al.* 2004).

Country Occurrence:

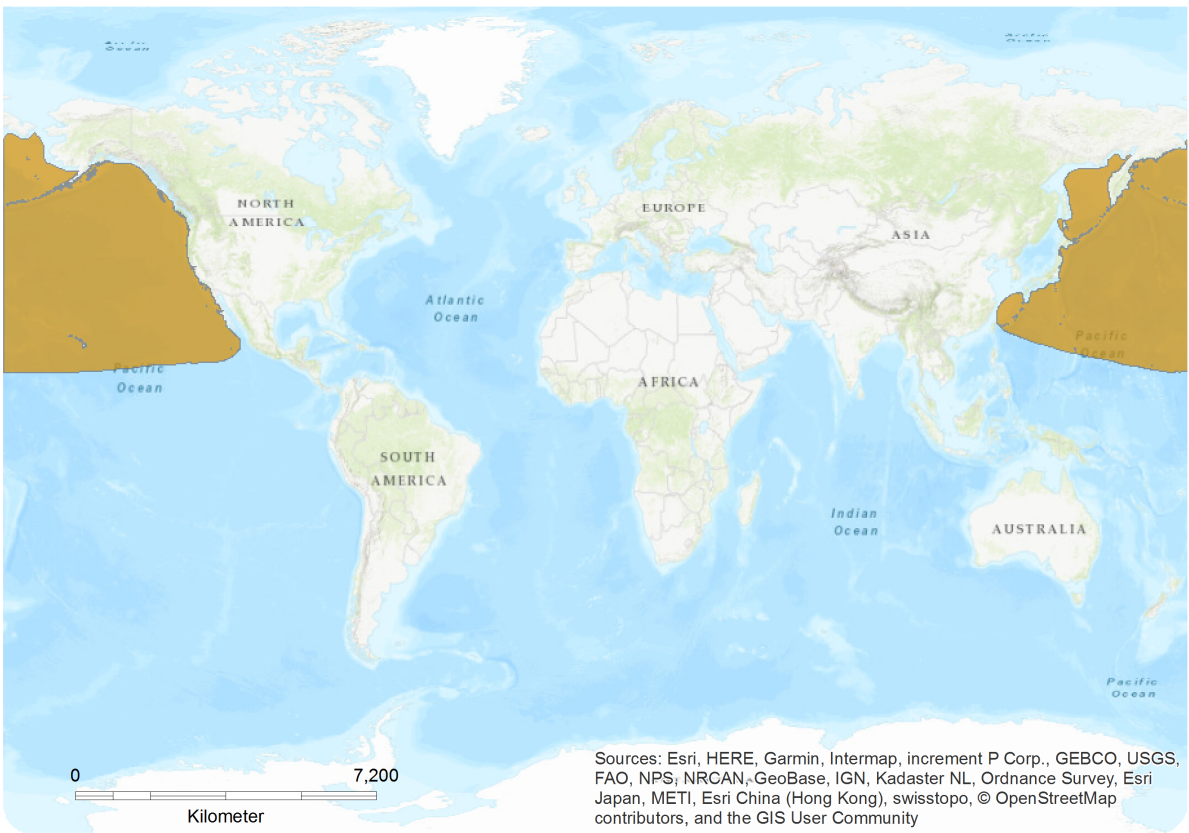
Native: Canada; Japan; Marshall Islands; Mexico; Micronesia, Federated States of ; Northern Mariana Islands; Russian Federation (Eastern Asian Russia); United States (Hawaiian Is.); United States Minor Outlying Islands

Vagrant: New Zealand; Norfolk Island

Present - origin uncertain: China; Korea, Republic of; Taiwan, Province of China

Distribution Map

Phoebastria immutabilis

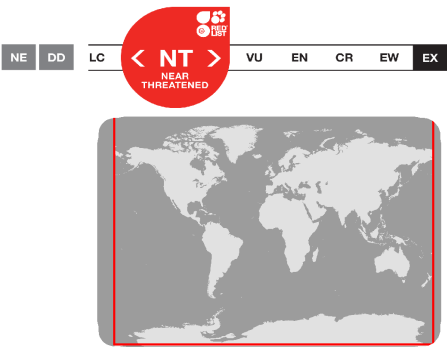


Range

- Extant (breeding)
- Extant (resident)

Compiled by:

BirdLife International and Handbook of the Birds of the World (2016)



Population

The population is estimated to be more than c. 800,000 breeding pairs, equivalent to c.1,600,000 mature individuals (Arata *et al.* 2009). Midway Atoll, Laysan Island and French Frigate Shoals support 90% of the global breeding population.

Trend Justification

Population sizes at monitored colonies increased between 1980 and 1995 but have never reached the densities observed prior to large-scale harvests for feathers in the early 1900s. Data indicate a 32% decline during 1992-2002 (3.2% per annum) in populations on Midway Atoll, Laysan Island and French Frigate Shoals in the Northwestern Hawaiian Islands, where 90% of the global population is found (Gilman and Freifeld 2003, USFWS data *per* B. Flint 2003). However, data from 2004 and 2005 indicate that the breeding population then rebounded, and that the overall population trend for 1992-2005 is stable, perhaps because these changes in the breeding populations were reflective of large scale environmental conditions that affected the number of birds that returned to the colonies to nest rather than actual declines in the population (Naughton *et al.* 2007). A population began nesting in Mexico in the 1980s and has been increasing since then. The current population is about 400 pairs at four sites, although this represents less than 0.1% of the global population (Naughton *et al.* 2007). Considering such data, and the difficulty of predicting long-term trends for such a long-lived species, a decline of 20-29% over the period 1992-2078 is precautionarily estimated given the number of documented threats and the uncertainty over their future effects.

Current Population Trend: Stable

Habitat and Ecology (see Appendix for additional information)

Laysan Albatross is an annual breeder though, like other albatross species, each year a proportion of birds will skip a breeding season. Nests vary from a simple scoop in the sand to more elaborate nests where vegetation allows. Wisdom, a *Phoebastria immutabilis* female estimated to be 65 years old is still breeding (https://www.fws.gov/refuge/Midway_Atoll/wildlife_and_habitat/Wisdom_Profile.html). High rates of mate change (14%), super-normal clutches as a result of same-sex pairing (31% of pairs), observed on Oahu and Kauai, Hawaii are all previously unreported for the species (Young *et al.* 2008, 2009a Young and VanderWerf 2014). The high rates of same-sex pairing is thought to result from a slightly skewed sex ratio (57% female) (Young *et al.* 2008). Diet analysis indicates that it feeds primarily on cephalopods (Pittman *et al.* 2004), though also on a variety of fish, crustaceans and other invertebrates (Tickell 2000). Satellite tracking has revealed the large journeys made even while breeding (Fernandez *et al.* 2001, Naughton *et al.* 2007, S. Shaffer *et al.* 2007). Breeding birds at Tern Island switch from a local unimodal foraging strategy during brooding, to a bimodal foraging strategy incorporating more distant, highly productive subarctic waters during the rearing period and responding to climatic variability readily (Hyrenbach *et al.* 2002, Kappes *et al.* 2010, Thorne *et al.* 2015). Reproductive success has been linked to foraging location, with pairs fledging chicks two years in a row not foraging near continental shelves (Edwards and Parrish 2008). Recent data has demonstrated that reproductive success is linked to the distance of the breeding grounds from the transition zone (closer resulting in higher reproductive success) and prevailing wind conditions in any given year (Thorne *et al.* 2015, 2016). It is thought that this ability may be influenced by information gathered during the non-

breeding season (Edwards and Parrish 2008).

Systems: Terrestrial, Marine

Threats (see Appendix for additional information)

Populations were greatly reduced by feather and egg collecting in the late 1800s and early 1900s, and by high seas drift nets for squid and salmon that were active between 1978 and 1992. Prior to its closure, the high seas driftnet fishery killed over 17,500 Laysan Albatross in 1990 alone (Johnson *et al.* 1993). Current key threats include incidental capture in pelagic (Crowder and Myers 2002) and demersal longline fisheries (Stehn *et al.* 2001) in the North Pacific as well as in illegal high seas driftnet operations. Analyses in 2001 estimated that pelagic longliners in the North Pacific may kill 5,000-18,000 *Phoebastria immutabilis* per year, with 8,000 thought the most likely figure, while demersal longline operations in the Bering Sea and Gulf of Alaska groundfish fisheries were estimated to kill c.715 birds per year (Crowder and Myers 2001). It has been recently estimated that on average, 156 *Phoebastria immutabilis* are taken annually in the Alaskan federal groundfish and Pacific Halibut *Hippoglossus stenolepis* fisheries, (in the Bering Sea and Aleutian Islands, and the Gulf of Alaska Fishery Management Plan areas), targeting Pacific Cod *Gadus macrocephalus*, Sablefish *Anoplopoma fimbria*, and Halibut *Hippoglossus* spp.. Off Canada's west coast, the estimated annual bycatch mortality of albatrosses (primarily Black-footed Albatross) averages 85 birds per year (range 25-128) (COSEWIC 2013, K. Morgan, *unpublished data*, 2016). The bycatch rates in Japanese and Taiwanese pelagic longline fisheries in the North Pacific are still largely unknown. Climate change poses an ongoing threat in terms of storms and flooding, and a future threat in terms of habitat alteration from sea-level rise. A majority of the population nests on low-lying islands and atolls and it is predicted that with climate change and the resulting rise in sea-level, there will be significant inundation and loss nesting habitat (Storlazzi *et al.* 2013, Reynolds *et al.* 2015). Other threats include pollution from oil spills, organochlorine, lead and plastic ingestion, as well as invasive species, human disturbance and conflicts with aircrafts (Harrison 1990, Ludwig *et al.* 1998, Finkelstein *et al.* 2003, Finkelstein 2006, Young *et al.* 2014). Chicks with large volumes of proventricular plastic have been reported to have fledging weights significantly lower than chicks with low amounts of plastic, and there is some evidence to suggest it may have affected survival rates in 1986, when the volume of plastic ingestion was at its highest (Sievert and Sileo 2008). Up to 10,000 chicks per year are potentially affected by lead poisoning from paint on buildings at Midway Atoll (Finkelstein 2006), and 7% of chicks on Sand Island fail to fledge from lead poisoning each year, with a predicted impact of 16% reduction in population size over the course of 50 years equating to 190,000 fewer birds (Finkelstein *et al.* 2010). Avian pox virus affects chicks on Midway and the Main Hawaiian Islands where introduced mosquitoes are present, but the associated declines are negligible (VanderWerf and Young 2016). Invasive predators also pose a threat. Dogs *Canis lupus familiaris* kill adults and chicks on inhabited islands in Hawaii. Observations from Guadalupe Island, Mexico, in December 2002 recorded that, of the 490 adults on the island, 35 were killed by feral cats *Felis catus* (7% of the total island breeding population that year, and 30% of the Punta Sur colony) (Keitt *et al.* 2006). Cats were also responsible for a tripling in nest failure rate at this colony (49%) compared with the nearby cat free Negro Islet colony (13%) (Keitt *et al.* 2006). Cat control around the colony from Jan-Mar 2003 removed 18 cats and curtailed adult mortality from cat predation, although cats continued to be seen around the colony after the breeding season ended (Keitt *et al.* 2006). It is likely that the population here had not experienced high levels of predation prior to 2002, as otherwise it would have been extirpated. Other introduced predators (notably the Polynesian rat *Rattus exulans*; Jones *et al.* 2008) may pose a minor threat to colonies in Mexico and on the Main Hawaiian Islands. Rats have been

recorded killing adults on rare occasions, but this mortality is minimal and hence not believed to be driving significant declines. House mice *Mus musculus* on Sand Island began attacking adult Laysan Albatross in 2015, with the behaviour spreading rapidly in the following breeding season. An eradication is planned and trapping around the colony has minimised chick losses. In 2002, observations on Clarion Island, Mexico, reported zero breeding success from the 46 nesting pairs, largely as a result of predation by several endemic species (e.g. Clarion Raven *Corvus corax clarionensis* and Clarion Racer *Masticophis anthonyi*; Wanless *et al.* 2009).

Conservation Actions (see Appendix for additional information)

Conservation Actions Underway

All of the major Hawaiian breeding localities are part of the US National Wildlife Refuge system or State of Hawaii Seabird Sanctuaries and, in 2006, the Papahānaumokuākea Marine National Monument was established, encompassing all of the Northwestern Hawaiian Islands. Originally, Papahānaumokuākea protected an area of 362,073 km²; in August 2016, the National Marine Monument was expanded to 1,508,870 km². Three breeding sites, supporting over 90% of the breeding population, are either counted directly or sampled at regular intervals. In 1991, a 50 Nautical Mile Protected Species Zone was established around the Northwestern Hawaiian Islands (primarily to protect monk seals). No longline fishing is allowed in this zone. In 2001 and 2007, respectively, USA and Canada released National Plans of Action to reduce the bycatch of seabirds in longline fisheries (NMFS 2001, DFO 2007). In 2002, the use of tori (streamer) lines became a mandatory condition of licence in commercial halibut, sablefish, and rockfish *Sebastes* spp. longline fisheries on Canada's west coast (DFO 2007). Awareness programmes and mitigation trials have been started in several major longline fleets operating within the foraging range of this species; for example, the Hawaiian longline fishing fleet is required to use measures to reduce bycatch of seabirds. In 2006, the Western and Central Pacific Fisheries Commission adopted a measure to require large longline vessels to use at least two seabird bycatch mitigation measures when fishing north of 23 degrees North. Predator control programs are conducted at colonies in Mexico and the main Hawaiian Islands (Young *et al.* 2013). **Conservation Actions Proposed**

Continue monitoring population trends and demographic parameters. Continue satellite-tracking studies to assess temporal and spatial overlap with longline fisheries. Adopt best-practice mitigation measures in longline fisheries within the species's range. Continue and enhance awareness programmes in all longline fleets. Re-evaluate the location of the current boundary (23° N) for required use of seabird mitigation measures in the U.S. pelagic longline fisheries (Naughton *et al.* 2007). Continue and enhance control/eradication programs for *Verbesina* in the Northwestern Hawaiian Islands and introduced predators in Hawaii and Mexico. Continue and enhance lead-based paint removal from sites (e.g. USFWS 2005-2008 efforts to remove paint from about 14 out of 95 structures with lead-based paint on Sand Island; Finkelstein *et al.* 2010).

Credits

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External Resources

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Appendix

Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Habitat	Season	Suitability	Major Importance?
9. Marine Neritic -> 9.1. Marine Neritic - Pelagic	Breeding	Suitable	Yes
9. Marine Neritic -> 9.1. Marine Neritic - Pelagic	Non-breeding	Suitable	Yes
9. Marine Neritic -> 9.2. Marine Neritic - Subtidal Rock and Rocky Reefs	Breeding	Suitable	No
9. Marine Neritic -> 9.2. Marine Neritic - Subtidal Rock and Rocky Reefs	Non-breeding	Suitable	No
9. Marine Neritic -> 9.3. Marine Neritic - Subtidal Loose Rock/pebble/gravel	Breeding	Suitable	No
9. Marine Neritic -> 9.3. Marine Neritic - Subtidal Loose Rock/pebble/gravel	Non-breeding	Suitable	No
9. Marine Neritic -> 9.4. Marine Neritic - Subtidal Sandy	Breeding	Suitable	No
9. Marine Neritic -> 9.4. Marine Neritic - Subtidal Sandy	Non-breeding	Suitable	No
9. Marine Neritic -> 9.5. Marine Neritic - Subtidal Sandy-Mud	Breeding	Suitable	No
9. Marine Neritic -> 9.5. Marine Neritic - Subtidal Sandy-Mud	Non-breeding	Suitable	No
9. Marine Neritic -> 9.7. Marine Neritic - Macroalgal/Kelp	Breeding	Suitable	No
9. Marine Neritic -> 9.7. Marine Neritic - Macroalgal/Kelp	Non-breeding	Suitable	No
9. Marine Neritic -> 9.9. Marine Neritic - Seagrass (Submerged)	Breeding	Suitable	No
9. Marine Neritic -> 9.9. Marine Neritic - Seagrass (Submerged)	Non-breeding	Suitable	No
10. Marine Oceanic -> 10.1. Marine Oceanic - Epipelagic (0-200m)	Breeding	Suitable	Yes
10. Marine Oceanic -> 10.1. Marine Oceanic - Epipelagic (0-200m)	Non-breeding	Suitable	Yes
10. Marine Oceanic -> 10.2. Marine Oceanic - Mesopelagic (200-1000m)	Breeding	Suitable	Yes
10. Marine Oceanic -> 10.2. Marine Oceanic - Mesopelagic (200-1000m)	Non-breeding	Suitable	Yes
10. Marine Oceanic -> 10.3. Marine Oceanic - Bathypelagic (1000-4000m)	Non-breeding	Suitable	Yes
10. Marine Oceanic -> 10.4. Marine Oceanic - Abyssopelagic (4000-6000m)	Non-breeding	Suitable	Yes
12. Marine Intertidal -> 12.1. Marine Intertidal - Rocky Shoreline	Breeding	Suitable	Yes
12. Marine Intertidal -> 12.2. Marine Intertidal - Sandy Shoreline and/or Beaches, Sand Bars, Spits, Etc	Breeding	Suitable	Yes

Habitat	Season	Suitability	Major Importance?
12. Marine Intertidal -> 12.3. Marine Intertidal - Shingle and/or Pebble Shoreline and/or Beaches	Breeding	Suitable	Yes

Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Threat	Timing	Scope	Severity	Impact Score
11. Climate change & severe weather -> 11.1. Habitat shifting & alteration	Future	Minority (50%)	Slow, significant declines	Low impact: 3
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		
11. Climate change & severe weather -> 11.4. Storms & flooding	Ongoing	Minority (50%)	Causing/could cause fluctuations	Low impact: 5
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		
5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.4. Unintentional effects: (large scale) [harvest]	Ongoing	Majority (50-90%)	Slow, significant declines	Medium impact: 6
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (<i>Felis catus</i>)	Ongoing	Minority (50%)	Negligible declines	Low impact: 4
	Stresses:	2. Species Stresses -> 2.1. Species mortality 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (<i>Rattus exulans</i>)	Ongoing	Minority (50%)	Negligible declines	Low impact: 4
	Stresses:	2. Species Stresses -> 2.1. Species mortality 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (<i>Mus musculus</i>)	Ongoing	Minority (50%)	Negligible declines	Low impact: 4
	Stresses:	2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (<i>Canis familiaris</i>)	Ongoing	Minority (50%)	Negligible declines	Low impact: 4
	Stresses:	2. Species Stresses -> 2.1. Species mortality 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		

9. Pollution -> 9.4. Garbage & solid waste	Ongoing	Majority (50-90%)	Unknown	Unknown
	Stresses:	2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		

Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Actions in Place
In-Place Research, Monitoring and Planning
Action Recovery plan: Yes
Systematic monitoring scheme: No
In-Place Land/Water Protection and Management
Conservation sites identified: Yes, over entire range
Occur in at least one PA: Yes
Invasive species control or prevention: No
In-Place Species Management
Successfully reintroduced or introduced benignly: No
Subject to ex-situ conservation: No
In-Place Education
Subject to recent education and awareness programmes: No
Included in international legislation: Yes
Subject to any international management/trade controls: No

Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Actions Needed
2. Land/water management -> 2.1. Site/area management
4. Education & awareness -> 4.3. Awareness & communications
5. Law & policy -> 5.1. Legislation -> 5.1.1. International level

Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Research Needed
1. Research -> 1.5. Threats

Research Needed
3. Monitoring -> 3.1. Population trends

Additional Data Fields

Distribution
Estimated area of occupancy (AOO) (km ²): 3500
Continuing decline in area of occupancy (AOO): Unknown
Extreme fluctuations in area of occupancy (AOO): No
Estimated extent of occurrence (EOO) (km ²): 49000000
Continuing decline in extent of occurrence (EOO): Unknown
Extreme fluctuations in extent of occurrence (EOO): No
Number of Locations: 16
Continuing decline in number of locations: Unknown
Extreme fluctuations in the number of locations: No
Lower elevation limit (m): 0
Upper elevation limit (m): 50
Population
Number of mature individuals: 1600000
Continuing decline of mature individuals: Yes
Extreme fluctuations: No
Population severely fragmented: No
No. of subpopulations: 3
Continuing decline in subpopulations: Unknown
Extreme fluctuations in subpopulations: No
All individuals in one subpopulation: No
Habitats and Ecology
Continuing decline in area, extent and/or quality of habitat: Unknown
Generation Length (years): 28.5
Movement patterns: Full Migrant
Congregatory: Congregatory (and dispersive)

The IUCN Red List Partnership



The IUCN Red List of Threatened Species™ is produced and managed by the [IUCN Global Species Programme](#), the [IUCN Species Survival Commission \(SSC\)](#) and [The IUCN Red List Partnership](#).

The IUCN Red List Partners are: [Arizona State University](#); [BirdLife International](#); [Botanic Gardens Conservation International](#); [Conservation International](#); [NatureServe](#); [Royal Botanic Gardens, Kew](#); [Sapienza University of Rome](#); [Texas A&M University](#); and [Zoological Society of London](#).